[RESUME START - COPY BELOW THIS LINE]

TOLBERT OKOTH ASUMBA Full Stack Developer | Software Engineer Email: osumba30@gmail.com | LinkedIn: linkedin.com/in/tolbert-asumba-542b3b344 | GitHub: github.com/Tolbert-Okoth

PROFESSIONAL SUMMARY Innovative Full Stack Developer and BSc Mathematics & Computer Science student with a strong foundation in algorithmic logic and system architecture. Proven track record of building complex, scalable web applications that integrate AI, Fintech APIs, and real-time data processing. Passionate about solving local problems through technology, from defense-grade telemedicine systems to agricultural marketplaces.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), Python, SQL, HTML5, CSS3.
- **Frontend:** React.js, Redux, Bootstrap 5, Tailwind CSS, Recharts, React-Leaflet.
- **Backend:** Node.js, Express, Python (FastAPI/Flask), Socket.io.
- **Database & Cloud:** PostgreSQL, MongoDB, Firebase, Cloudinary, Render, Vercel.
- **Tools & AI:** Git, M-Pesa API, Groq AI (Llama-3), PyTorch, Torchaudio, Alpaca Finance API.

FEATURED PROJECTS

SautiYaAfya | AI-Powered Pediatric Respiratory Triage *Lead Developer | Ministry of Health (Kenya)*

- Developed a defense-grade telemedicine platform for diagnosing pediatric pneumonia and asthma using AI audio analysis.
- Engineered a hybrid AI engine combining CNNs (MobileNetV2), DSP (Mel-Spectrograms), and Groq LLM for interpretable diagnostics.
- Implemented an offline-first architecture to support Community Health Workers in remote areas with auto-sync capabilities.
- Created a real-time epidemiology surveillance map to track high-risk disease clusters across counties.

Afya-Pulse | AI-Driven Healthcare Triage System *Full Stack Developer*

- Architected a real-time medical surveillance platform designed to streamline patient intake and prioritize critical cases.
- Developed a microservices architecture connecting a React frontend with a Python-based AI Engine.
- Integrated Groq Llama-3 to analyze symptoms (supporting Sheng & Swahili) and categorize medical urgency.
- Implemented secure Service-to-Service authentication and Token Refreshing using Firebase to ensure data security.

Ghoul System | Autonomous Market Surveillance Tool *Lead Developer*

- Built a "Cyberpunk" tactical dashboard for real-time financial market analysis and autonomous trading signal generation.
- Engineered a dual-service backend (Manager + Brain) to process live financial news via WebSockets and Alpaca API.
- Utilized technical indicators (TA-Lib) and LLMs to generate automated trading strategies in real-time.

FarmConnect | Multi-Vendor Agricultural Marketplace *Full Stack Developer*

- Developed a comprehensive e-commerce platform connecting small-scale farmers directly to buyers.
- Successfully integrated the M-Pesa API (STK Push & B2C) to facilitate secure mobile payments and wallet withdrawals.
- Created a multi-vendor admin dashboard for inventory management and refund processing with secure JWT authentication.

EDUCATION & CERTIFICATIONS

BSc Mathematics & Computer Science *Current Student* Focus: Applied Mathematics, Algorithms, Data Structures, and Software Engineering.

freeCodeCamp Certifications

- **Data Analysis with Python** (Jan 2026)
- **Scientific Computing with Python** (Oct 2025)
- **Quality Assurance (Node.js)** (Aug 2025)
- **Relational Database (PostgreSQL)** (Jul 2025)
- **Back End Development and APIs** (Jul 2025)
- **Data Visualization** (Jun 2025)
- **Front End Development Libraries** (Jun 2025)
- **JavaScript Algorithms and Data Structures** (Jun 2025)
- **Responsive Web Design** (Jun 2025)